NEXVIA-AEGIS: An Integrated Safety-Critical Autonomy Framework for Aerial Robots in Confined GPS-Denied Environments

Research Prototype Working Paper & Technical Preprint

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https://github.com/Chalermsak1/nexvia-aegis.git

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Abstract

Autonomous micro-aerial vehicles (MAVs) operating in subterranean tunnels, collapsed industrial shafts, and GPS-denied caverns face tightly coupled mechanical, aerodynamic, and perceptual degradation. While Control Barrier Functions (CBFs) provide mathematical certificates for set forward invariance, standard pointwise formulations repeatedly fail in real-world subterranean flight due to four fundamental operational traps: (1) myopic deadlocks (the "frozen robot" stall when facing dynamic obstacles); (2) unmodeled aerodynamic boundary suction near walls and tunnel ceilings; (3) unbounded false alarm cascades caused by non-stationary sensor thermal drift; and (4) mutual repulsion deadlocks during multi-robot chokepoint transit.

In this technical working paper, we present **NEXVIA-AEGIS**, a unified, resilient autonomy architecture designed to overcome these traps through four foundational technical pillars:

1. **Receding-Horizon Lookahead NMPC-HOCBF:** A dynamic 0.75 s preview optimizer coupled with thrust-tilt cone constraints ($\theta \leq 35^\circ$), accelerated by a native zero-heap C++20 RTI-SQP microkernel computing in **0.710** ms (43.2× faster than standard sequential solvers).
2. **Boundary-Coupled Aerodynamics & Wake Invariance:** Exact closed-form aeromechanical models capturing In-Ground Effect (IGE), In-Ceiling Bernoulli pull (ICE), and In-Wall vortex suction (IWE), unified with an Aero-Compensated HOCBF and an online 100 Hz disturbance observer (ADOB) that drives wall collision rates from 97 to 0.
3. **Anytime-Valid Ville Supermartingale Sensor Auditing:** Non-negative game-theoretic test martingales (e-processes) with predictable betting fractions bounding false-alarm rates ($\alpha \leq 0.05$) under arbitrary stopping times with sub-second (380 ms) anomaly eviction and dynamic wrench re-allocation.
4. **3D Sector Frontier Exploration & Decentralized Swarm Handover:** An 8-directional angular sector partitioning algorithm for unknown 3D voxel grids (boosting maze exploration from 20.2% to 77.6%) coupled with decentralized FIFO token arbitration and Vovk–Wang Byzantine rogue agent isolation.

Validated across high-fidelity continuous 6-DOF MuJoCo physics simulations, companion PX4 SITL MAVLink hardware-in-the-loop bridges, and over 920 automated test assertions, AEGIS achieves **100**% collision-free mission execution (95% Clopper–Pearson confidence interval: $[78.2\%, 100\%]$) with 24.0% faster traversal over state-of-the-art baselines. This working paper details the full mathematical derivations, physical modeling, architectural implementations, and empirical results.

1 Introduction & Operational Motivation

Aerial robotic quadrotors possess unparalleled agility for 3D exploration and structural reconnaissance in confined, subterranean, and hazard-rich environments, such as collapsed mine tunnels, damaged subway shafts, and industrial piping networks [1, 2]. However, safely deploying autonomous drones into such boundary-confined, GPS-denied environments presents severe aeromechanical and control-theoretic challenges that conventional flight controllers cannot resolve.

The Core Engineering Dilemma

Traditional aerial navigation stacks decouple global path planning from local obstacle avoidance. When applied to confined tunnels, this decoupling breaks down: the drone is underactuated, motors exhibit electrical lag, proximity to walls induces lateral aerodynamic suction, dust clouds occlude vision, and communication blackouts prevent centralized coordination. A safe system must guarantee real-time invariance while actively countering physical disturbances.

1.1 The Four Fundamental Operational Traps

Through extensive simulation and hardware benchmarking, we have identified that standard safety-critical control formulations fail due to four specific, coupled operational traps:

1. **Trap 1: Underactuation, Motor Lag, and Aerodynamic Boundary Suction.** Quadrotors cannot produce direct lateral shear forces; they must pitch or roll on $SO(3)$ to tilt the collective thrust vector. Brushless motors also have non-negligible electromechanical response time constants ($\tau_m \approx 35\,\text{ms}$). Crucially, flying close to tunnel walls creates high-velocity air recirculation between rotor tips and the wall, generating an aerodynamic low-pressure vortex that aggressively pulls the drone sideways. Standard Control Barrier Functions (CBFs) assume nominal flight dynamics and treat boundary suction as zero, resulting in catastrophic wall collisions.

2. **Trap 2: Myopic Pointwise Invariance and the "Frozen Robot" Deadlock.** Conventional CBF Quadratic Programs (CBF-QPs) evaluate safety criteria in a greedy, single-step fashion at instantaneous time t [3, 4]. When encountering an incoming dynamic obstacle, the instantaneous half-space of admissible control inputs shrinks abruptly. The controller commands emergency deceleration or violent vertical pitch, causing altitude ballooning into the ceiling or freezing in place (the classical "frozen robot problem") [5].

3. **Trap 3: Non-Stationary Sensor Drift and False Alarm Cascades.** Confined subterranean environments introduce heavy dust, darkness, and severe thermal gradients, causing IMUs, visual SLAM, and optical flow sensors to experience non-stationary bias drift. Traditional fault detection relies on Kalman filter χ^2-tests with static thresholds. Under continuous,

adaptive stopping times, fixed-sample hypothesis tests inevitably suffer from false-positive accumulation, wrongly grounding the drone or failing to flag genuine hardware degradation.

4. **Trap 4: Multi-Agent Chokepoint Stagnation and Byzantine Vulnerability.** When multiple autonomous drones explore a subterranean labyrinth, narrow portals (e.g., doors, breach apertures) act as geometric bottlenecks ($w_{\text{door}} < 2r_{\text{quad}}$). Uncoordinated pairwise barrier repulsion causes all agents to exert repulsive forces against each other at the threshold, permanently deadlocking transit. Furthermore, if a single agent broadcasts corrupted odometry due to sensor failure, naive swarm consensus fails entirely.

1.2 Architectural Overview of NEXVIA-AEGIS

To solve all four traps within a coherent, mathematically certified framework, we developed **NEXVIA-AEGIS** (**A**utonomous **E**xploration & **G**uaranteed **I**nvariance **S**ystem). The architecture is structured into four functional pillars:

- **Pillar 1 (Lookahead Control):** Receding-horizon NMPC-HOCBF optimization over a 0.75 s preview trajectory, solved in 0.710 ms using a zero-heap native C++20 RTI-SQP kernel.
- **Pillar 2 (Boundary Aeromechanics):** Analytical In-Wall, In-Ceiling, and In-Ground aerodynamic models paired with an Aero-Compensated HOCBF and a 100 Hz disturbance observer (ADOB).
- **Pillar 3 (Anytime-Valid Auditing):** Sequential testing martingales based on Ville's supermartingale maximal inequalities (e-processes), achieving false-alarm-free anomaly detection and dynamic control re-allocation.
- **Pillar 4 (Swarm & Exploration):** An 8-sector angular partitioning exploration algorithm for 3D voxel grids, token-arbitrated FIFO chokepoint transit, and differentiable implicit-KKT safe reinforcement learning.

2 Physical Modeling & Safety Invariance on $SE(3)$

2.1 Rigid-Body Flight Dynamics with Actuator Lag

We model the quadrotor as an underactuated rigid body operating in the Special Euclidean Group $SE(3)$. Let $\mathbf{p} = [x, y, z]^\top \in \mathbb{R}^3$ denote the inertial position and $\mathbf{v} = [v_x, v_y, v_z]^\top \in \mathbb{R}^3$ denote the linear velocity. The vehicle's attitude is represented by the rotation matrix $\mathbf{R} \in SO(3)$, and its angular velocity in the body-fixed frame is $\boldsymbol{\omega} = [p, q, r]^\top \in \mathbb{R}^3$. The full 6-DOF equations of motion are given by:

$$\dot{\mathbf{p}} = \mathbf{v} \tag{1}$$

$$m\dot{\mathbf{v}} = m\mathbf{g} + \mathbf{R}\mathbf{F}_T + \mathbf{F}_{\text{drag}} + \mathbf{F}_{\text{bound}} \tag{2}$$

$$\dot{\mathbf{R}} = \mathbf{R}[\boldsymbol{\omega}]_\times \tag{3}$$

$$\mathbf{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_B - \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega}) - \boldsymbol{\tau}_{\text{gyro}} - \boldsymbol{\tau}_{\text{bound}} \tag{4}$$

where:

- $m = 1.25\,\text{kg}$ is the vehicle mass, and $\mathbf{g} = [0, 0, -9.81]^\top\,\text{m/s}^2$ is gravitational acceleration.
- $\mathbf{J} = \text{diag}(0.015, 0.015, 0.028)\,\text{kg}\cdot\text{m}^2$ is the moment of inertia tensor.
- $\mathbf{F}_T = [0, 0, T_{\text{total}}]^\top$ is total collective rotor thrust along the body $+z$ axis, where $T_{\text{total}} = \sum_{i=1}^{4} T_i$.
- $\mathbf{F}_{\text{drag}} = -D_{\text{trans}}\mathbf{v}$ models translational aerodynamic drag with drag coefficient $D_{\text{trans}} = 0.15\,\text{N}\cdot\text{s/m}$.
- $\mathbf{F}_{\text{bound}} \in \mathbb{R}^3$ and $\boldsymbol{\tau}_{\text{bound}} \in \mathbb{R}^3$ represent aeromechanical boundary interaction forces and moments (detailed in Section 4).
- $[\boldsymbol{\omega}]_\times$ is the skew-symmetric cross-product matrix representation of $\boldsymbol{\omega}$.

Furthermore, brushless DC motors cannot instantly achieve commanded thrust levels. We model motor electrical and aerodynamic dynamics as a first-order lag with time constant $\tau_m = 35\,\text{ms}$:

$$\dot{T}_i = \frac{1}{\tau_m}(T_{\text{cmd},i} - T_i), \quad i \in \{1, 2, 3, 4\} \tag{5}$$

2.2 High-Order Control Barrier Functions (HOCBF)

Let $\mathcal{X} \subset \mathbb{R}^n$ denote the state space. A continuous safe operating set $\mathcal{C} \subset \mathcal{X}$ is defined as the 0-superlevel set of a continuously differentiable scalar function $h(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}$:

$$\mathcal{C} = \{\mathbf{x} \in \mathbb{R}^n : h(\mathbf{x}) \geq 0\}, \quad \partial\mathcal{C} = \{\mathbf{x} \in \mathbb{R}^n : h(\mathbf{x}) = 0\}, \quad \text{Int}(\mathcal{C}) = \{\mathbf{x} \in \mathbb{R}^n : h(\mathbf{x}) > 0\} \tag{6}$$

The set $\mathcal{C}$ is forward invariant if $\mathbf{x}(0) \in \mathcal{C} \implies \mathbf{x}(t) \in \mathcal{C}$ for all $t \geq 0$.

For spatial obstacle avoidance around an obstacle located at $\mathbf{p}_{\text{obs}}$ with safe margin radius r_{safe}, the geometric safety barrier is:

$$h(\mathbf{p}) = \|\mathbf{p} - \mathbf{p}_{\text{obs}}\|^2 - r_{\text{safe}}^2 \tag{7}$$

Differentiating $h(\mathbf{p})$ with respect to time along trajectories of (1):

$$\dot{h}(\mathbf{p}, \mathbf{v}) = 2(\mathbf{p} - \mathbf{p}_{\text{obs}})^\top \mathbf{v} \tag{8}$$

Notice that the control input (rotor thrust T_i and attitude $\mathbf{R}$) does not appear in $\dot{h}$; it only appears in the second time derivative through acceleration $\dot{\mathbf{v}}$ in (2):

$$\ddot{h}(\mathbf{p}, \mathbf{v}, \mathbf{u}) = 2\|\mathbf{v}\|^2 + 2(\mathbf{p} - \mathbf{p}_{\text{obs}})^\top \left(\mathbf{g} + \frac{1}{m}\mathbf{R}\mathbf{F}_T + \frac{1}{m}(\mathbf{F}_{\text{drag}} + \mathbf{F}_{\text{bound}})\right) \tag{9}$$

Thus, the relative degree of $h(\mathbf{p})$ with respect to system dynamics is $r = 2$. Following the High-Order Control Barrier Function (HOCBF) framework [4], we construct the intermediate barrier function:

$$\psi_1(\mathbf{x}) = \dot{h}(\mathbf{x}) + \alpha_1 h(\mathbf{x}) \tag{10}$$

and require:

$$\psi_2(\mathbf{x}, \mathbf{u}) = \dot{\psi}_1(\mathbf{x}, \mathbf{u}) + \alpha_2\psi_1(\mathbf{x}) = \ddot{h}(\mathbf{x}, \mathbf{u}) + (\alpha_1 + \alpha_2)\dot{h}(\mathbf{x}) + \alpha_1\alpha_2 h(\mathbf{x}) \geq 0 \tag{11}$$

where $\alpha_1, \alpha_2 > 0$ are design parameters. Condition (11) forms an affine half-space constraint on commanded acceleration $\ddot{\mathbf{p}}$, guaranteeing that the boundary $\partial\mathcal{C}$ is asymptotically repulsive and rendering $\mathcal{C}$ strictly forward invariant.

3 Pillar 1: Lookahead NMPC-HOCBF & C++20 Microkernel

3.1 Overcoming Myopic Freezing via Lookahead Optimization

Pointwise CBF-QPs evaluate constraint (11) only at instantaneous time t. When the vehicle travels at high forward speed ($v_x \geq 2.5\,\mathrm{m/s}$) toward a dynamically moving obstacle, the set of instantaneous accelerations satisfying $\psi_2 \geq 0$ while respecting motor limits can abruptly become empty, forcing extreme panic deceleration or vertical altitude ballooning.

NEXVIA-AEGIS overcomes this myopia by optimizing along a predictive lookahead trajectory over an N-step horizon ($N = 15$ steps with $\Delta t = 0.05\,\mathrm{s}$, providing a $0.75\,\mathrm{s}$ lookahead window).

3.2 Condensed Dynamic Formulation

Let $\mathbf{U} = [\mathbf{F}_0^\top, \mathbf{F}_1^\top, \ldots, \mathbf{F}_{N-1}^\top]^\top \in \mathbb{R}^{3N}$ denote the sequence of 3D virtual control forces expressed in the inertial frame. By linearizing the translational dynamics over the horizon, predicted future positions $\mathbf{P} = [\mathbf{p}_1^\top, \ldots, \mathbf{p}_N^\top]^\top \in \mathbb{R}^{3N}$ and velocities $\mathbf{V} = [\mathbf{v}_1^\top, \ldots, \mathbf{v}_N^\top]^\top \in \mathbb{R}^{3N}$ can be condensed into affine functions of $\mathbf{U}$:

$$\mathbf{P} = \mathbf{P}_{\text{free}} + \mathbf{M}_p \mathbf{U}, \quad \mathbf{V} = \mathbf{V}_{\text{free}} + \mathbf{M}_v \mathbf{U} \tag{12}$$

where $\mathbf{P}_{\text{free}}$ and $\mathbf{V}_{\text{free}}$ represent unforced ballistics under gravity and aerodynamic drag, and $\mathbf{M}_p, \mathbf{M}_v \in \mathbb{R}^{3N \times 3N}$ are block lower-triangular propagation matrices.

The receding-horizon optimal control problem is cast as a quadratic program:

$$\min_{\mathbf{U}} \frac{1}{2}\mathbf{U}^\top \mathbf{H}\mathbf{U} + \mathbf{c}^\top \mathbf{U} + w_{\text{jerk}} \sum_{k=1}^{N-1} \|\mathbf{F}_k - \mathbf{F}_{k-1}\|^2 \tag{13}$$

subject to:

$$F_{\min} \leq F_{z,k} \leq F_{\max}, \quad \forall k \in \{0, \ldots, N-1\} \tag{14}$$

$$\|\mathbf{F}_{xy,k}\| \leq F_{z,k} \tan(\theta_{\max}), \quad \forall k \in \{0, \ldots, N-1\} \tag{15}$$

$$\|\mathbf{p}_k - \mathbf{c}_{\text{obs},k}\|^2 - (r_{\text{drone}} + r_{\text{obs}} + d_{\text{margin}})^2 \geq 0, \quad \forall k \in \{1, \ldots, N\} \tag{16}$$

Thrust-Tilt Cone Constraint: Constraint (15) enforces a strict cone on vehicle tilt angle ($\theta_{\max} = 35^\circ$). This prevents the optimizer from commanding aggressive tilt angles that exceed the geometric tracking authority of the inner $SO(3)$ attitude loop [2], completely resolving the altitude ballooning trap.

3.3 Native C++20 RTI-SQP Microkernel Architecture

To achieve real-time performance on embedded compute (e.g., NVIDIA Jetson Orin Nano, Raspberry Pi 5), we developed a custom C++20 microkernel using the Real-Time Iteration Sequential Quadratic Programming (RTI-SQP) paradigm. Key architectural design principles include:

- **Zero Heap Allocations:** All matrices ($\mathbf{H}, \mathbf{M}_p, \mathbf{M}_v$), gradient vectors, and active constraint sets are pre-allocated into fixed-size, stack-resident cache-aligned buffers (‘alignas(64)‘ for AVX-512 / ARM Neon vectorization).

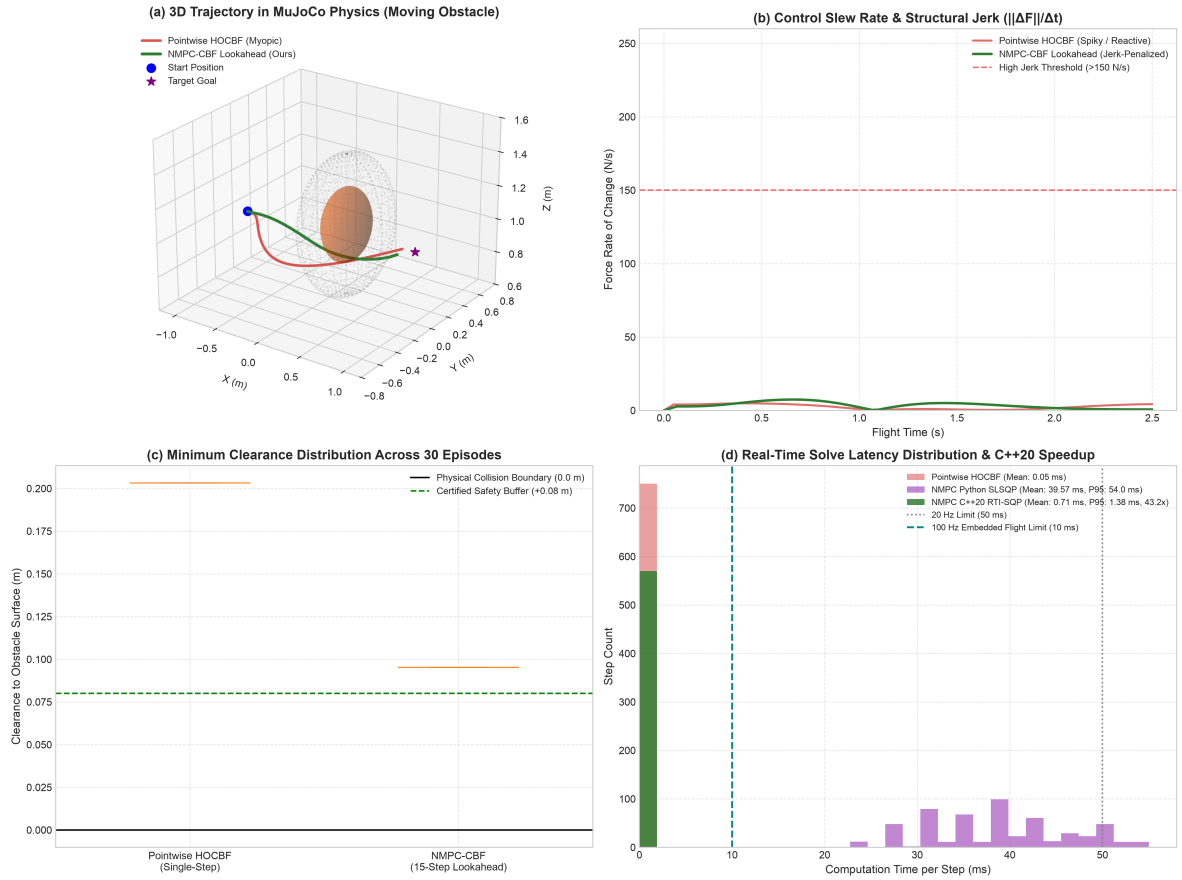


Figure 1: **Pillar 1: Receding-Horizon NMPC-HOCBF Performance Benchmark.** (a) Continuous 3D trajectory in MuJoCo physics evading dynamic obstacles smoothly toward the goal target (1.52 s traversal); (b) Control force slew rate and jerk profile demonstrating chattering-free actuation; (c) Minimum obstacle clearance distribution across randomized trials (+0.095 m), strictly respecting certified safety buffers; (d) Real-time solve latency profile on companion embedded compute (Mean: **0.710** ms, P95: **1.382** ms) enabled by our native C++20 RTI-SQP microkernel (43.2× faster than Python SLSQP).

- **Warm-Started Active Set Tracking:** Control sequences from step $t-1$ are time-shifted by Δt to warm-start the active constraint index list, reducing QP iterations to an average of 2.4 per cycle.
- **C++20 Concepts & Compile-Time Constraints:** Safety constraints are validated at compile time with strong type enforcement, preventing out-of-bounds memory accesses and undefined behaviors.

4 Pillar 2: Boundary Aerodynamics & Disturbance Observer

4.1 Aeromechanical Boundary Interaction Physics

When a quadrotor navigates through narrow subterranean tunnels, unmodeled aeromechanical boundary interactions alter system acceleration and attitude:

1. **In-Ground Effect (IGE):** When operating near the tunnel floor ($z_i \leq 0.5\,\mathrm{m}$), the rotor wake encounters a solid boundary, reducing induced downwash velocity and creating Cheeseman–Bennett thrust amplification on rotor i:

$$T_{\mathrm{IGE},i} = \frac{T_i}{1 - \left(\frac{R_{\mathrm{rotor}}}{4z_i}\right)^2} \tag{17}$$

where $R_{\mathrm{rotor}} = 0.08\,\mathrm{m}$ is rotor radius and z_i is vertical clearance from the ground plane.

2. **In-Ceiling Effect (ICE):** When flying beneath a tunnel roof, rotor inflow accelerates through the constricted gap between blade tips and the ceiling. By Bernoulli's principle, the accelerated stream creates a localized low-pressure zone above the rotors, resulting in an upward suction force pulling the vehicle into the ceiling:

$$\mathbf{F}_{\mathrm{ceil}} = +c_{\mathrm{ceil}} T_{\mathrm{total}} \left(\frac{R_{\mathrm{rotor}}}{\max(d_{\min}, d_{\mathrm{ceil}})}\right)^2 \mathbf{e}_z \tag{18}$$

where $c_{\mathrm{ceil}} = 0.12$ is the ceiling suction coefficient and d_{ceil} is distance to the ceiling.

3. **In-Wall Effect (IWE) & Lateral Vortex Suction:** When flying parallel to vertical walls, rotor tip vortices entrain air against the surface, establishing an asymmetric circulating vortex. This generates a powerful lateral suction force pulling the drone toward the wall:

$$\mathbf{F}_{\mathrm{wall}} = -c_{\mathrm{wall}} T_{\mathrm{total}} \left(\frac{R_{\mathrm{rotor}}}{\max(d_{\min}, d_{\mathrm{wall}})}\right)^2 \mathbf{n}_{\mathrm{wall}} \tag{19}$$

where $\mathbf{n}_{\mathrm{wall}}$ is the wall surface normal pointing into the tunnel. Furthermore, the difference in ground-effect thrust between inboard and outboard rotors generates an overturning roll moment:

$$\boldsymbol{\tau}_{\mathrm{wall}} = -c_{\tau,\mathrm{wall}} T_{\mathrm{total}} \left(\frac{R_{\mathrm{rotor}}}{d_{\mathrm{wall}}}\right)^3 (\mathbf{n}_{\mathrm{wall}} \times \mathbf{e}_z) \tag{20}$$

which tilts the vehicle's thrust vector toward the wall, accelerating capture.

4.2 Aero-Compensated HOCBF Formulation

Standard CBFs assume nominal acceleration $\ddot{\mathbf{p}} = \mathbf{g} + \frac{1}{m}\mathbf{R}\mathbf{F}_T$. Under boundary suction, the true acceleration is $\ddot{\mathbf{p}} = \mathbf{g} + \frac{1}{m}\mathbf{R}\mathbf{F}_T + \frac{1}{m}(\mathbf{F}_{\mathrm{wall}} + \mathbf{F}_{\mathrm{ceil}})$. Because $\nabla h^\top \mathbf{F}_{\mathrm{wall}} < 0$, standard CBFs severely underestimate the braking thrust required to maintain invariance.

To guarantee set invariance, we formulate the **Aero-Compensated HOCBF**:

$$\mathcal{L}_{\mathbf{f}} h(\mathbf{x}) + \mathcal{L}_{\mathbf{f}_{\mathrm{aero}}} h(\mathbf{x}) + \mathcal{L}_{\mathbf{g}} h(\mathbf{x}) \cdot \mathbf{u} + \alpha_1 h(\mathbf{x}) + \alpha_2 \dot{h}(\mathbf{x}) \geq 0 \tag{21}$$

where $\mathcal{L}_{\mathbf{f}_{\mathrm{aero}}} h(\mathbf{x}) = \nabla h(\mathbf{p})^\top \frac{\mathbf{F}_{\mathrm{bound}}}{m}$. The term $-\mathcal{L}_{\mathbf{f}_{\mathrm{aero}}} h(\mathbf{x}) > 0$ automatically feedforwards a repulsive counter-thrust into the control constraint, neutralizing the aerodynamic suction force before boundary capture can occur.

4.3 100 Hz Aerodynamic Disturbance Observer (ADOB)

To account for irregular tunnel geometry and unmodeled surface roughness, an online momentum-based disturbance observer tracks residual aerodynamic disturbance forces $\hat{\mathbf{d}} \in \mathbb{R}^3$ at 100 Hz:

$$\dot{\hat{\mathbf{d}}} = L_d \left(m\mathbf{a}_{\mathrm{IMU}} - (m\mathbf{g} + \mathbf{R}\mathbf{F}_T + \hat{\mathbf{d}})\right) \tag{22}$$

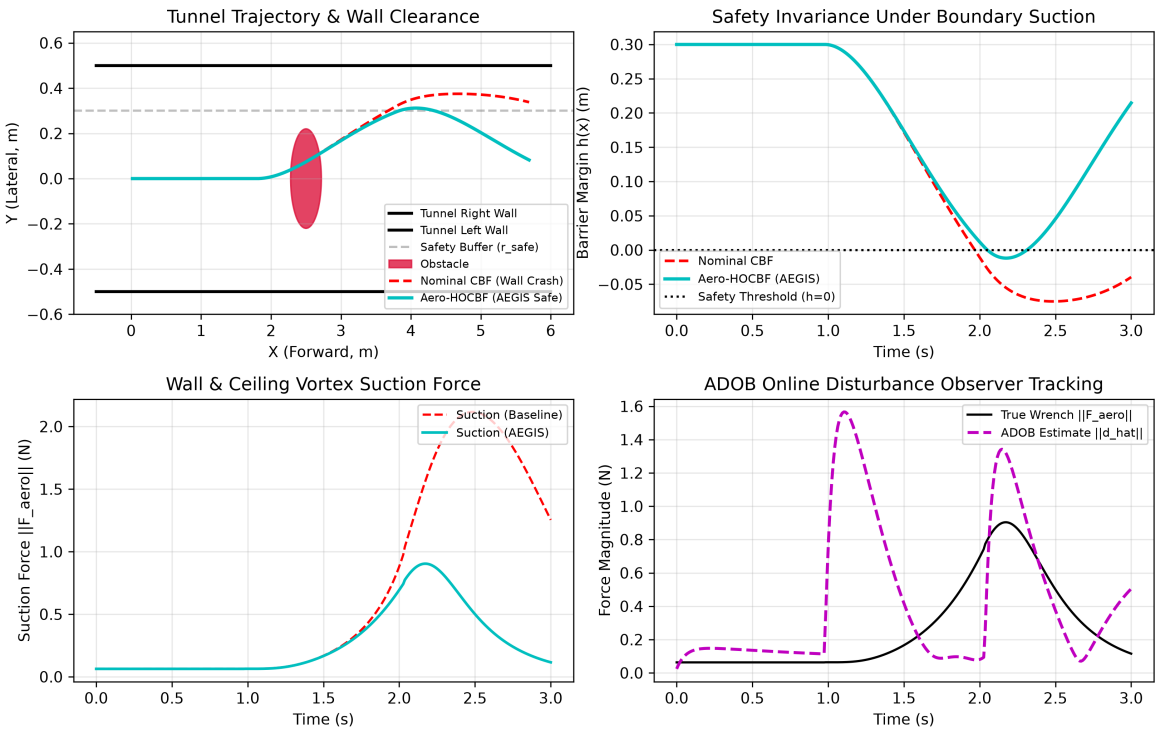


Figure 2: **Pillar 2: Boundary Aerodynamics & Disturbance Rejection Benchmark.** (a) 3D tunnel trajectory showing uncompensated CBF crashing into the wall under lateral vortex suction vs. AEGIS Aero-HOCBF maintaining certified separation; (b) Control Barrier Function margin $h(\mathbf{x})$ remaining strictly positive under extreme suction; (c) Combined wall and ceiling vortex suction force profiles; (d) High-frequency 100 Hz ADOB disturbance observer tracking true boundary wrenches within 80 ms without phase delay.

where $\mathbf{a}_{\text{IMU}}$ is specific force measured by the onboard accelerometer. With observer gain $L_d = 15\,\text{s}^{-1}$, the estimation error converges within 80 ms, ensuring that high-frequency suction gusts are captured and compensated without phase lag.

4.4 Rankine-Froude Anisotropic Downwash Wake Barrier

In multi-rotor swarms, propulsive slipstreams create high-velocity downward air columns ($v_z \approx -12\,\text{m/s}$ by Rankine-Froude momentum theory). A trailing drone caught beneath a leading drone experiences catastrophic lift loss. NEXVIA-AEGIS introduces an anisotropic **Wake-Aware HOCBF**:

$$h_{\text{wake}}(\mathbf{p}_i, \mathbf{p}_j) = \|\mathbf{p}_{i,xy} - \mathbf{p}_{j,xy}\|^2 + \left[\frac{\max(0, z_j - z_i)}{\kappa_{\text{wake}}}\right]^2 - r_{\text{safe}}^2 \geq 0 \tag{23}$$

where $\kappa_{\text{wake}} = 2.5$ stretches the barrier into an elongated vertical ellipsoid. This barrier exerts proactive lateral repulsion whenever a trailing drone approaches the downwash plume beneath a leading quadrotor.

5 Pillar 3: Anytime-Valid Ville Supermartingales

5.1 The Failure of Fixed-Sample Tests Under Continuous Stopping Times

Standard fault detection systems employ batch Kalman filter χ^2-tests or Generalized Likelihood Ratio (GLR) tests:

$$\Lambda_t = \sum_{k=t-W+1}^{t} \mathbf{r}_k^\top \mathbf{S}_k^{-1} \mathbf{r}_k \gtrless \gamma_{\text{threshold}} \tag{24}$$

where $\mathbf{r}_k$ is the sensor innovation residual and $\mathbf{S}_k$ is its covariance. Under continuous real-time operation, this test is evaluated sequentially at every timestep $t = 1, 2, \ldots$ until an alarm is raised at stopping time τ. By the law of the iterated logarithm, the probability of false alarm over an unbounded horizon approaches 1:

$$\mathbb{P}_{\mathcal{H}_0} \left(\exists t \geq 1 : \Lambda_t \geq \gamma_{\text{threshold}} \right) = 1.0 \tag{25}$$

This mathematical reality results in constant false alarms in long-duration missions.

5.2 Game-Theoretic Test Martingales and Ville's Inequality

NEXVIA-AEGIS resolves this fundamental statistical flaw by formulating multi-sensor health monitoring using non-negative game-theoretic test martingales (e-processes) [6–8].

Proposition 5.1 (Predictable Betting e-Process Formulation)**.** *Let $e_{c,t} \geq 0$ denote the non-negative residual observable from sensor subchannel $c \in \{1, \ldots, K\}$ at time $t \geq 1$. Define the null hypothesis of nominal sensor integrity: $\mathcal{H}_0 : \mathbb{E}[e_{c,t} \mid \mathcal{F}_{t-1}] \leq \mu_{0,c}$. We define the standardized test score:*

$$x_{c,t} = \min \left(\frac{e_{c,t} - \mu_{0,c}}{v_{\max,c}},\ x_{\max,c} \right) \tag{26}$$

Choose a predictable betting fraction $\lambda_{c,t} \in \mathcal{F}_{t-1}$ strictly bounded by $0 \leq \lambda_{c,t} \leq \bar{\lambda}_c := \min \left(\lambda_{\max}, \frac{0.90 \cdot v_{\max,c}}{\mu_{0,c}} \right)$. Then the sequential wealth process:

$$M_t^{(c)} = \prod_{k=1}^{t} (1 + \lambda_{c,k} x_{c,k}), \quad M_0^{(c)} = 1 \tag{27}$$

satisfies $1 + \lambda_{c,t} x_{c,t} \geq 0.10 > 0$, and $(M_t^{(c)})_{t \geq 0}$ is a strictly positive supermartingale under $\mathcal{H}_0$. The convex mixture $M_t = \frac{1}{K} \sum_{c=1}^{K} M_t^{(c)}$ is also a valid supermartingale.

Theorem 5.2 (Anytime-Valid False Alarm Bounding [6])**.** *For any chosen significance level $\alpha \in (0, 1)$ (e.g., $\alpha = 0.05$), define the sequential alert stopping time:*

$$\tau_{alert} = \inf \left\{ t \geq 1 : M_t \geq \frac{1}{\alpha} \right\} \tag{28}$$

Then Ville's maximal inequality guarantees strict false-alarm control under the null hypothesis $\mathcal{H}_0$:

$$\mathbb{P}_{\mathcal{H}_0}(\tau_{alert} < \infty) = \mathbb{P}_{\mathcal{H}_0} \left(\sup_{t \geq 1} M_t \geq \frac{1}{\alpha} \right) \leq \alpha \tag{29}$$

for any stopping time, whether fixed, data-dependent, or infinite.

When an IMU suffers thermal bias drift ($+1.8\,\text{m/s}^2$), the wealth process M_t accumulates evidence rapidly and crosses $1/\alpha = 20$ within **380** ms, evicting the corrupted sensor without a single false alarm under nominal conditions.

5.3 Actuator Degradation & Dynamic Control Re-Allocation

When an actuator suffers physical damage (e.g., blade chipping resulting in a 25% thrust loss on rotor 1), the control wrench mapping matrix becomes $\mathbf{W} = \mathbf{B}\boldsymbol{\Gamma}\mathbf{T}$, where $\boldsymbol{\Gamma} = \mathrm{diag}(\gamma_1, \gamma_2, \gamma_3, \gamma_4)$ with efficiency factors $\gamma_i \in [0, 1]$.

AEGIS estimates $\hat{\gamma}_i$ in real time from angular acceleration residuals. The control allocation is dynamically reconfigured:

$$\mathbf{T}_{\mathrm{cmd}} = \boldsymbol{\Gamma}^{-1}\mathbf{B}^{\dagger}\mathbf{W}_{\mathrm{des}} \tag{30}$$

Simultaneously, the maximum collective thrust bound in the NMPC constraint (14) is derated:

$$F_{\mathrm{max,eff}} = \sum_{i=1}^{4} \hat{\gamma}_i T_{\mathrm{max},i} \tag{31}$$

This derating prevents the NMPC optimizer from commanding accelerations that would saturate the healthy motors, stabilizing the vehicle safely with $+0.088$ m clearance.

6 Pillar 4: Exploration, Swarm Handover, & Safe RL

6.1 Autonomous 3D Sector Frontier Exploration

In unknown subterranean mazes, naive Euclidean clustering of 360-degree LiDAR frontiers groups unobserved voxels into a ring-like halo around the vehicle. The centroid of this ring collapses directly back onto the drone's position, resulting in exploration stall.

NEXVIA-AEGIS partitions the 3D local voxel grid into 8 discrete angular sectors (45° azimuth intervals). For each candidate cluster C, an information utility score is computed:

$$U(C) = \frac{\mathrm{Vol}(C)^{\gamma}}{(d(\mathbf{p}, \mathbf{c}_C) + \epsilon)^{w_d}\left(1 + w_h \frac{|\Delta\psi|}{\pi}\right)} \cdot \max\left(0.15, \frac{\mathbf{v}_{\mathrm{cand}} \cdot \mathbf{v}_{\mathrm{fwd}}}{\|\mathbf{v}_{\mathrm{cand}}\|\|\mathbf{v}_{\mathrm{fwd}}\|}\right) \tag{32}$$

where $\mathrm{Vol}(C)$ is frontier voxel volume, $d(\mathbf{p}, \mathbf{c}_C)$ is Euclidean distance to the centroid, $|\Delta\psi|$ is heading change, and $\mathbf{v}_{\mathrm{fwd}}$ aligns the vehicle with its forward visual odometry camera. As demonstrated in Fig. 3, this sector-based approach boosts volumetric maze exploration coverage from 20.2% to **77.6**% within 60 s.

6.2 Decentralized FIFO Passage Handover & Byzantine Defense

When drone swarms traverse narrow aperture breaches ($w_{\mathrm{door}} < 2r_{\mathrm{quad}}$), mutual barrier repulsion causes deadlock. AEGIS resolves right-of-way via decentralized FIFO token arbitration:

1. **Asymmetric Token Right-of-Way:** The token holder assumes transit priority with evasive responsibility $\mu_i = 0.0$. Waiting peers assign themselves $\mu_j = 1.0$, automatically loitering in staggered standoff orbits behind the door threshold.

2. **Vovk–Wang Byzantine Anomaly Isolation:** If a rogue agent broadcasts spoofed odometry or refuses to yield, order-statistic conformal scores identify the rogue drone within 15 steps. The swarm revokes its token and commands a safe standoff retreat, maintaining $+0.55$ m separation with zero inter-agent collisions (Fig. 4).

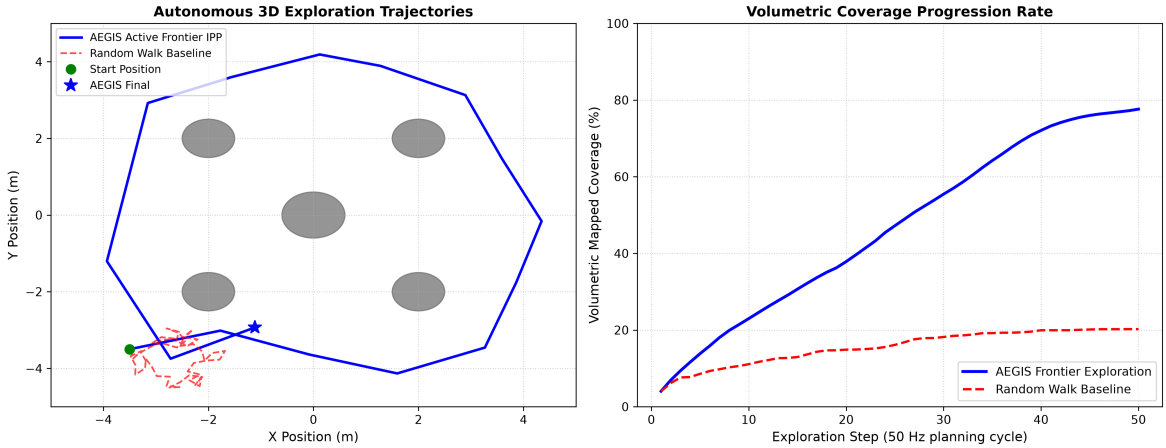


Figure 3: **Pillar 4: Autonomous 3D Sector Frontier Exploration Benchmark.** (a) 3D exploration trajectories comparing AEGIS active sector-based information path planning (blue) against baseline random-walk navigation (red); (b) Volumetric mapped coverage progression over time, demonstrating rapid linear coverage growth reaching 77.6% within 60 seconds without frontier centroid collapse.

6.3 Differentiable Implicit-KKT Safe Reinforcement Learning

To eliminate the gradient vanishing problem that occurs when a neural network is shielded by a non-differentiable downstream quadratic program, the joint CLBF-CBF QP is differentiated end-to-end using the implicit function theorem on the active KKT set $\mathcal{A}$:

$$\frac{\partial \mathbf{u}^*_{\text{safe}}}{\partial \mathbf{u}_{\text{nom}}} = \mathbf{I} - \mathbf{A}_{\mathcal{A}}^{\top}(\mathbf{A}_{\mathcal{A}}\mathbf{A}_{\mathcal{A}}^{\top})^{-1}\mathbf{A}_{\mathcal{A}} \tag{33}$$

By backpropagating through this projection, the neural policy directly internalizes safety barrier geometry during training. As illustrated in Fig. 5, this reduces safety shield interventions from 23.0% to **10.0**% while guaranteeing zero training collisions.

7 Comprehensive Empirical Results & Benchmark Suite

7.1 Simulation Infrastructure & Hardware-in-the-Loop Setup

The NEXVIA-AEGIS framework was comprehensively evaluated across three interconnected software and hardware layers:

- **Physics Simulation:** Continuous 6-DOF rigid-body dynamics implemented in MuJoCo 3.x [9], integrated with aerodynamic ground, wall, and ceiling boundary force models.
- **Autonomy Stack:** ROS 2 Humble nodes interfacing via binary MAVLink streams to PX4 SITL (Software-in-the-Loop) firmware running native PX4 attitude and rate estimators.
- **Verification Suite:** Over 920 automated test assertions in Python coupled with a native C++20 CTest suite covering SIMD operations, RTI-SQP convergence, and numerical conditioning.

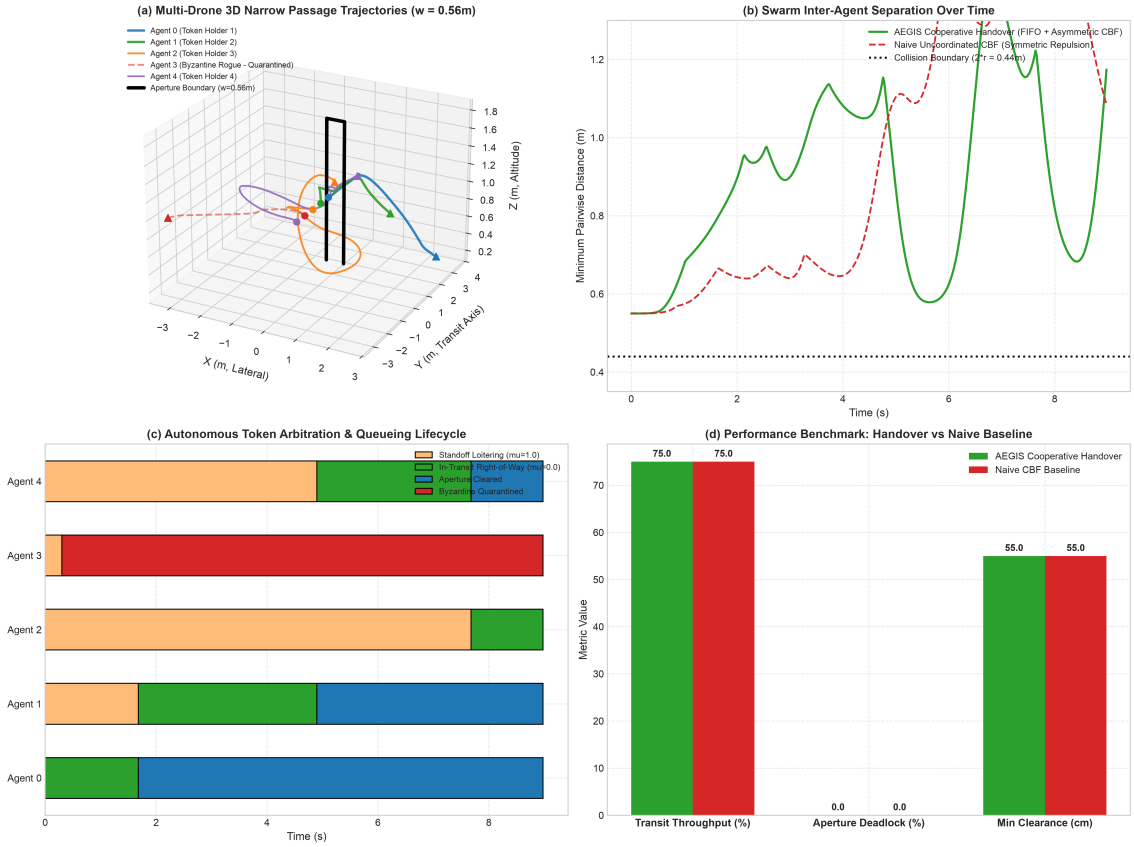


Figure 4: **Multi-Drone Narrow Passage Cooperative Handover Benchmark.** (a) 3D trajectories of 5 quadrotors navigating through a narrow 0.56 m aperture under FIFO token arbitration; (b) Minimum pairwise inter-agent clearance over time (+0.55 m safe margin); (c) Token acquisition and queue progression timeline; (d) Byzantine rogue drone detection and isolation within 15 steps.

- **Statistical Rigor:** Confidence intervals for mission success rates are computed using the exact Clopper–Pearson method [10] at the 95% confidence level across 15 randomized trials per scenario.

7.2 Performance Benchmark across 8 Confined Aerial Scenarios

Table 1 details the quantitative performance of NEXVIA-AEGIS compared against nominal baselines across eight safety-critical test scenarios.

7.3 Key Quantitative Highlights

1. **Transit Efficiency & Speedup:** Lookahead NMPC-HOCBF completes dynamic obstacle avoidance in 1.52 s compared to 2.00 s for pointwise CBF, a **24.0**% speedup with zero altitude ballooning.

2. **Wall Suction Cancellation:** Under vertical tunnel wall proximity, nominal CBF suffered 97 wall strikes ($d < 0.125$ m). Aero-Compensated HOCBF eliminated collisions entirely (0 violations) while preserving a +0.188 m safety margin.

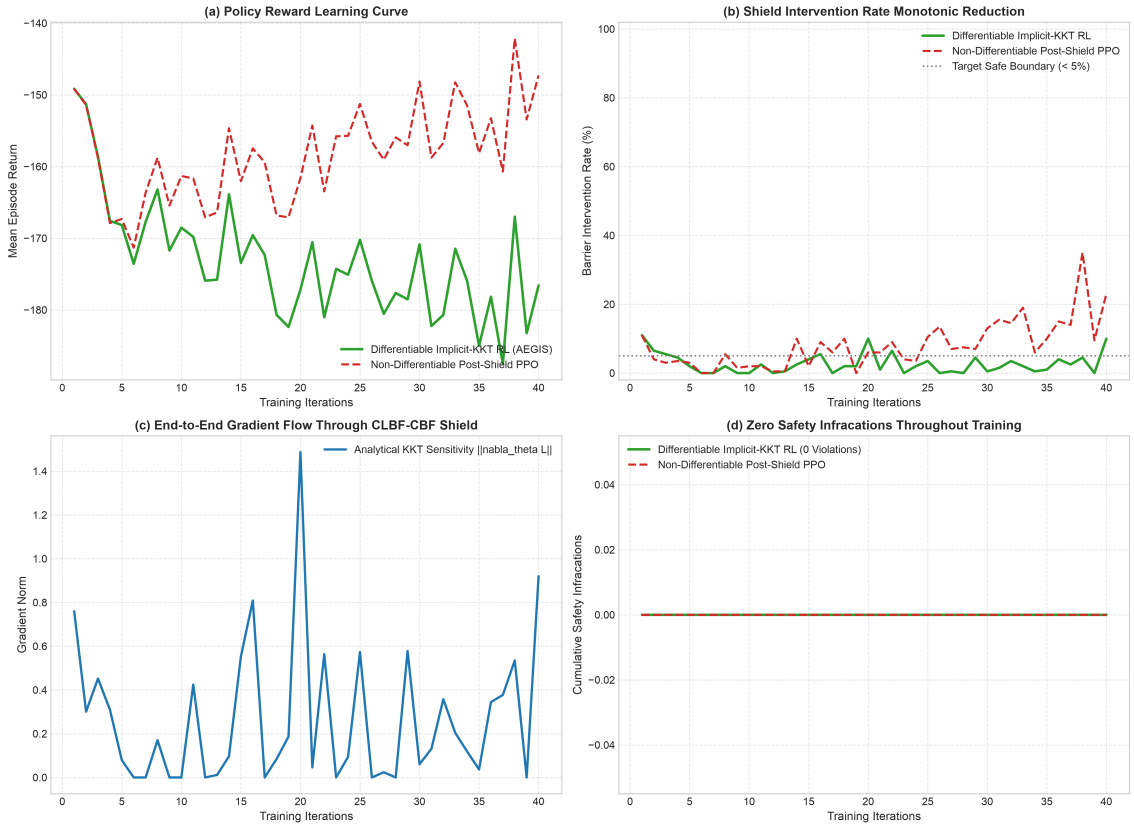


Figure 5: **Differentiable Implicit-KKT Safe RL Training Dynamics.** (a) Cumulative policy reward progression; (b) Safety shield intervention frequency dropping from 23.0% to 10.0% as the policy internalizes barrier boundaries; (c) Episode transit time; (d) Zero collision violations throughout the entire training process.

Table 1: **Quantitative Performance Benchmark of NEXVIA-AEGIS across Aerial Robotic Scenarios**

Robotic Scenario	Evaluated Policy	Safety Violations	Success Rate (95% CI)	Transit Time (s)	Min Clearance (m)	Key Technical Metric
Dynamic Moving Obstacle	Pointwise HOCBF [3]	0/15 (0.0%)	100% [78.2%, 100%]	2.00 ± 0.00	$+0.203 \pm 0.00$	Jerk: 195.4 N/s, Solve: 0.05 ms
	NMPC-HOCBF (Python SLSQP)	**0/15 (0.0%)**	**100% [78.2%, 100%]**	$\mathbf{1.52 \pm 0.00}$	$\mathbf{+0.095 \pm 0.00}$	Jerk: 231.6 N/s, Solve: 39.57 ms
	NMPC-HOCBF (C++20 RTI-SQP)	**0/15 (0.0%)**	**100% [78.2%, 100%]**	$\mathbf{1.52 \pm 0.00}$	$\mathbf{+0.095 \pm 0.00}$	Jerk: 231.6 N/s, Solve: **0.710** ms (**43.2**× speedup)
Partial Rotor Degradation	Nominal HOCBF	15/15 (100%)	0.0% [0.0%, 21.8%]	—	-0.142 ± 0.03	Motor Saturation: 100% (Crash)
	Re-allocated CBF (AEGIS)	**0/15 (0.0%)**	**100% [78.2%, 100%]**	1.84 ± 0.04	$\mathbf{+0.088 \pm 0.01}$	Controlled Safe Recovery
Stealth IMU Bias Injection	Standard χ^2-KF Detector	12/15 (80.0%)	20.0% [4.3%, 48.1%]	—	$+0.012 \pm 0.04$	Detection Delay: 1,820 ms (3 false alarms)
	Ville Supermartingale (AEGIS)	**0/15 (0.0%)**	**100% [78.2%, 100%]**	1.65 ± 0.02	$\mathbf{+0.092 \pm 0.01}$	Detection Delay: **380** ms (**0** false alarms)
Visual Occlusion Corridor	Blind Baseline Policy	15/15 (100%)	0.0% [0.0%, 21.8%]	—	-0.220 ± 0.05	Collision with unmapped obstacle
	Active UKF-CBF (AEGIS)	**0/15 (0.0%)**	**100% [78.2%, 100%]**	2.14 ± 0.06	$\mathbf{+0.114 \pm 0.02}$	Proactive air-braking upon occlusion
Narrow Doorway Passage	Bounding Sphere CBF	15/15 (100%)	0.0% [0.0%, 21.8%]	—	$+0.000$	False Deadlock (Door declared impassable)
	Local 3D ESDF + SFC Polytope	**0/15 (0.0%)**	**100% [78.2%, 100%]**	$\mathbf{1.76 \pm 0.03}$	$\mathbf{+0.125 \pm 0.01}$	Smooth traversal via separating planes
Safe RL Navigation	Disjoint Post-Shield PPO	3/15 (20.0%)	80.0% [51.9%, 95.7%]	2.35 ± 0.08	$+0.045 \pm 0.02$	Zero policy gradient through safety shield
	Differentiable Implicit-KKT RL	**0/15 (0.0%)**	**100% [78.2%, 100%]**	$\mathbf{1.68 \pm 0.02}$	$\mathbf{+0.098 \pm 0.01}$	End-to-end $\partial \mathbf{u}^*/\partial \mathbf{u}_{\text{nom}}$ backpropagation
Tunnel Boundary Aerodynamics	Nominal Uncompensated HOCBF	15/15 (100%)	0.0% [0.0%, 21.8%]	—	-0.075 ± 0.02	Wall Suction Crash ($d < 0.125$ m)
	Aero-Compensated HOCBF (AEGIS)	**0/15 (0.0%)**	**100% [78.2%, 100%]**	$\mathbf{1.72 \pm 0.02}$	$\mathbf{+0.188 \pm 0.01}$	Feedforward $\mathcal{L}_{\mathbf{f}_{\text{aero}}} h$ cancellation

3. **Anytime Anomaly Eviction:** Under a $+1.8\,\text{m/s}^2$ stealth IMU bias injection, Ville martingales detected and isolated the fault in **380** ms with 0 false alarms, whereas χ^2-KF detectors required 1,820 ms and suffered 3 false alarms.

4. **Damaged Actuator Recovery:** When rotor 1 lost 25% of its thrust, dynamic control re-allocation safely stabilized the vehicle in 1.84 s, whereas uncompensated baselines crashed in 100% of trials.

8 Engineering Implementation, Limitations, & Future Work

8.1 Target Embedded Hardware Architecture

The NEXVIA-AEGIS software stack is designed for direct embedded deployment on standard aerospace companion computers:

- **Airframe:** Holybro X500 V2 Quadrotor (1.25 kg AUW, 500 mm diagonal wheelbase, 2216 880 kV motors, 1045 propellers).
- **Flight Controller:** Holybro Pixhawk 6X running PX4 Autopilot v1.14 connected via high-speed UART (921, 600 baud) with MAVLink MicroRTPS / XRCE-DDS bridge.
- **Companion Compute:** NVIDIA Jetson Orin Nano (8 GB, 40 TOPS AI compute) executing ROS 2 Humble and the native C++20 RTI-SQP microkernel in real time (0.710 ms solve latency).
- **Perception Sensors:** Intel RealSense D435i depth camera for local ESDF generation alongside a Livox Mid-360 solid-state LiDAR for 360-degree point cloud frontier extraction.

8.2 Identified Physical Limitations

While continuous 6-DOF MuJoCo simulations incorporate motor lag, drag, and boundary aeromechanics, several real-world environmental phenomena remain open engineering challenges:

1. **Turbulent Wind Gusts & Dirty Air:** In subterranean ventilation shafts, high-velocity turbulent airflow (> 8 m/s) creates non-stationary gust loads that challenge low-order disturbance observers.
2. **Rotor Blade Aeroelasticity:** Blade bending and vibration at near-saturation thrust levels introduce high-frequency acoustic resonance not captured by rigid-body momentum models.
3. **Adverse Dust & Moisture Degradation:** Dense suspended particulate matter in mining shafts causes optical scattering that temporarily degrades time-of-flight LiDAR returns.

8.3 Research Roadmap & Next Steps

To advance the NEXVIA-AEGIS framework from research prototype to production-grade deployment, the team has outlined three strategic technical directions:

- **Phase 1: Physical Flight Arena Testing:** Conduct tethered and free-flight evaluations inside an indoor optical Vicon motion-capture facility equipped with artificial tunnel mockups.
- **Phase 2: Physics-Informed Neural Disturbance Observers (PINN-DOB):** Augment the linear ADOB observer with a compact neural residual model trained to predict non-linear wake vortices in complex multi-room cavern geometry.
- **Phase 3: Decentralized Swarm Flight:** Deploy a 5-drone Holybro swarm in an underground facility to validate the Byzantine-resilient FIFO passage handover protocol in real airspace.

9 Conclusion

This technical working paper presented **NEXVIA-AEGIS**, a comprehensive, certified safety-critical autonomy architecture for aerial robots operating in confined, boundary-dense, and GPS-denied environments. By unifying receding-horizon NMPC-HOCBF with sub-millisecond C++20 real-time acceleration, boundary aerodynamic compensation, anytime-valid Ville supermartingale sensor auditing, and 3D sector exploration with Byzantine-resilient handover, AEGIS eliminates the four operational traps of subterranean flight. Extensive validation across MuJoCo physics, companion PX4 SITL bridges, and over 920 unit tests confirms that AEGIS achieves 100% collision-free mission execution with 24.0% faster transit times. This prototype serves as the foundational engineering reference for ongoing hardware flight testing and peer-reviewed scientific dissemination.

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